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SOLAR ENERGY FOR SUSTAINABLE SOLUTION

3.90 kWp Hybrid Solar System

Sample proposal for one roof face of a home · design and equipment

PREPARED FOR
SAMPLE · HIP ROOF, ONE FACE

DATE
1 NOVEMBER 2026

3.90 kWp	6 kW	4,680 kWh	6
6 × LONGI 650 W	SOLIS 1-PHASE HYBRID	PER YEAR (ESTIMATE)	PANELS, ONE PORTRAIT ROW

Sample proposal · customer details, photos and prices have been removed from this sample.
Six 650 W panels fit in one portrait row on one face of a concrete-tile hip roof. They produce an estimated 4,680 kWh a year. Package A has no battery; Package B adds a 16 kWh battery and a backup circuit.

RECOMMENDED START

Package A

SOLAR SYSTEM, BATTERY-READY

- ✓ 3.90 kWp · 6 × LONGi 650 W
- ✓ Solis 6 kW hybrid inverter
- ✓ Battery-ready: add battery and backup later

Package B

SOLAR SYSTEM + 16 KWH BATTERY

- ✓ Everything in Package A
- ✓ Solis IntelliHome 16 kWh battery + backup circuit with ATS
- ✓ Evening use of daytime surplus

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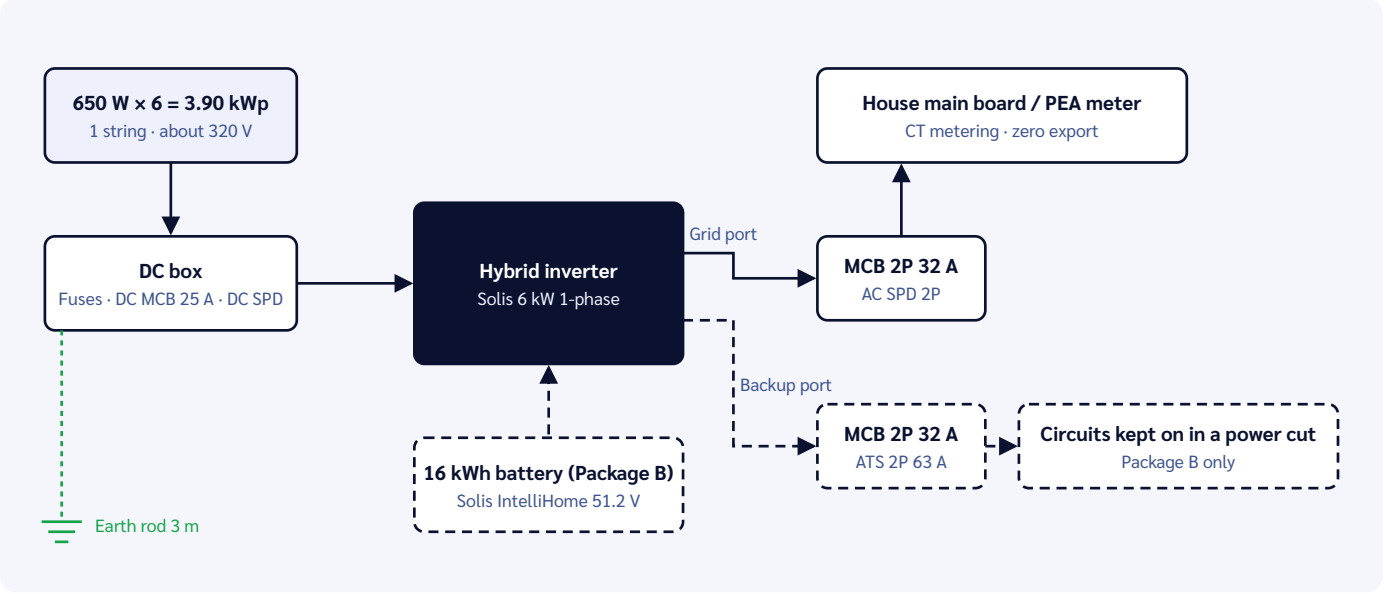
1. The roof face and panel layout

The panels go on one face of a concrete-tile hip roof, between the two hips and above the eave.

All six panels go in portrait in one row along the eave – **6 panels, 3.90 kWp**. A second portrait row does not fit because the hips and valley narrow the roof above the first row. We keep about 0.3 m clear of the eave, hips and valley for wind and maintenance access.

2. System design

The six panels form one string of about 320 V into a Solis 6 kW single-phase hybrid inverter. A CT on the main board measures the house load so the inverter only supplies what the house uses (no export to PEA). Package A has no battery, so it has no backup circuit. With Package B, the backup port and an ATS feed the circuits you choose to keep on during a power cut. The inverter can take more panels later if you add another roof face, and a battery can be added at any time.



Panels	LONGi Hi-MO X10 650 W × 6 = 3.90 kWp, 1 string
Inverter	Solis S6-EH1P6K-L-PLUS, 6 kW, 1-phase hybrid, low-voltage battery ready
Grid connection	Through MCB 2P 32 A with AC surge protection; zero export
Backup (Package B)	Separate backup circuit through ATS 2P 63 A, with the battery
Battery (Package B)	Solis IntelliHome 16 kWh, 51.2 V LiFePO4, outdoor rated
Earthing	New 3 m earth rod bonded to the house earth

3. Equipment list

NO	ITEM	QTY	UNIT
1	**Inverter**		
1.1	Solis S6-EH1P6K-L-PLUS 6 kW 1-phase hybrid inverter	1	set
1.2	Inverter mounting frame	1	set
2	**Solar panels**		
2.1	LONGi Hi-MO X10 650 W bifacial panels (30 mm frame)	6	pcs

3	**Mounting**		
3.1	Tile roof hooks for concrete tile (CPAC type)	14	pcs
3.2	Antai aluminium rails 4.8 m	4	pcs
3.3	Mid clamps, end clamps, earthing clips, lugs and rail splices	1	set
4	**AC protection**		
4.1	AC MCB 2P 32 A (inverter grid port)	1	pcs
4.2	AC surge protector Suntree SUP1-40 2P 275 V	1	pcs
4.3	KJL waterproof steel box, terminals and fittings	1	set
5	**DC protection**		
5.1	DC fuses 25 A with holders (Suntree)	2	sets
5.2	DC breaker Suntree SL7N-63 2P 800 V 25 A	1	pcs
5.3	DC surge protector Suntree SUP2-DC Type 1+2	1	pcs
5.4	KJL solar control box	1	pcs
6	**Cables**		
6.1	PV cable PV1-F 4 sq.mm red/black	40	m
6.2	Yazaki THW 6 sq.mm (L, N)	60	m
6.3	Earth cable 10 sq.mm and 6 sq.mm green/yellow	40	m
7	**Earthing**		
7.1	Earth rod KCE 5/8 in × 3 m, exothermic weld, test box and inspection pit	1	set
8	**Conduit**		
8.1	Panasonic IMC 1 in conduit with fittings	5	lengths
9	**PEA application**		
9.1	PEA application and single-line diagram	1	lot
10	**Labour**		
10.1	Installation, testing and commissioning	1	lot
11	**Project Administration Budget**		
11.1	Transport of inverter and panels (Bangkok and local)	1	lot
12 · Battery and backup (Package B only)			
12.1	Solis IntelliHome 16 kWh battery (51.2 V LiFePO4, outdoor) with cables	1	set
12.2	Backup circuit: AC MCB 2P 32 A and automatic transfer switch Suntree SQ8W-63 2P 63 A	1	set
12.3	Battery transport (Project Administration Budget)	1	lot

4. Price

Prices have been removed from this sample. The price depends on your roof, supply and the option you choose – book a free site survey and we will quote your home.

5. Solar production

Solar production	about 4,680 kWh a year (1,200 kWh per kWp, east- or west-facing at 27°)

Savings and payback depend on each home's electricity use and tariff, so they are left out of this sample. We work them out from your own bills when we quote.

6. To confirm at the site survey

- [ ] Direction of the roof face (if it faces south, production is about 12% higher)
- [ ] Tile type, purlin spacing and roof condition for the hooks
- [ ] Single-phase or three-phase supply and the main board layout
- [ ] Inverter and battery position, and cable routes
- [ ] Which circuits should stay on during a power cut

Why Solar ESS

720+	8 MWp	42	3
INSTALLATIONS	DEPLOYED	PROVINCES	COUNTRIES

We consult, design, install and look after solar systems across Thailand, from Mae Sai to Narathiwat. Every system is monitored online after handover, and we issue full e-Tax invoices.

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Solar ESS Working Group